Q: 1

Working with Files and JSON in Python

You are helping a small bookstore save and manage their inventory. The store currently has a list of books stored in a file called `inventory.json`. Your job is to read the inventory, make changes, and save it back — all using proper Python file-handling practices.

You are given the following starter file `inventory.json`:

```
[
  {"title": "The Alchemist", "author": "Paulo Coelho", "price": 12.99, "in_stock":
true},
  {"title": "1984", "author": "George Orwell", "price": 9.99, "in_stock": true}
]
```

And the following new book to add:

```
new_book = {"title": "Atomic Habits", "author": "James Clear", "price": 14.99,
"in_stock": True}
```

Task 1 — Read the inventory Open `inventory.json` using a `with` block and load its contents into a variable called `inventory`. Print the total number of books currently in the file.

Task 2 — Update and save Append `new_book` to the `inventory` list. Write the updated list back to `inventory.json` using a `with` block, with an indentation of 4 spaces.

Task 3 — Display the inventory Read the updated file again and print each book's details in the following format:

```
Title: The Alchemist | Author: Paulo Coelho | Price: $12.99
Title: 1984 | Author: George Orwell | Price: $9.99
Title: Atomic Habits | Author: James Clear | Price: $14.99
```

Submission Guidelines

- Write your code along with explanations in a Word file or Google Doc.
- Submit using Google Drive or GitHub only.

Google Drive:

- Set access to “Anyone with the link – Viewer.”
- Ensure the link opens without access requests.

GitHub:

- Upload files to a public repository.
- Ensure all required files are pushed and accessible.

Private links or inaccessible submissions will not be evaluated.